

ID	Paper (year)	Core method (very short)	Datasets / tasks (from text)	Baselines reported in text	Reported results vs baselines
1	Wang et al., HBM 2023 – UCGL: “Unsupervised contrastive graph learning for resting-state fMRI analysis and brain disorder detection” [hbm.26469]	Unsupervised contrastive graph learning framework (UCGL). Pretext stage: bi-level augmentation of BOLD time series, build FC graphs, two parallel GCNs trained with contrastive loss; downstream stage: fine-tune on labeled data for disease classification. arxiv	Three rs-fMRI datasets; cross-site and cross-dataset classification for three disorders: MDD, ASD, and AD (binary disease vs HC tasks). arxiv	Abstract only states “several state-of-the-art approaches”; specific method names are not visible in the accessible part of the article. arxiv	UCGL outperforms prior SOTA rs-fMRI methods for automated diagnosis of MDD, ASD, and AD in cross-site and cross-dataset settings; exact margins and baselines are not given in the accessible text. arxiv
2	Wang et al., MedIA 2024/25 – GCDA: “Self-supervised graph contrastive learning with diffusion augmentation for functional MRI analysis and brain disorder detection” [GCDA]	Self-supervised graph contrastive learning with diffusion-based graph augmentation (GCDA). Pretext model: construct FC graph from rs-fMRI, apply graph diffusion augmentation (diffusion model + Graph Transformer denoiser) on nodes/edges, then two shared-parameter GINs trained with SimSiam-style contrastive loss; task-specific model: fine-tune GIN on labeled target site with cross-entropy. pmc.ncbi.nlm.nih+1	REST-meta-MDD (three largest sites; MDD vs HC) and ABIDE (NYU, UM, LEUVEN; ASD vs HC). Cross-site training: pretrain on largest site, fine-tune + test on other sites. pmc.ncbi.nlm.nih	12 baselines: non-learnable augmentations on BOLD/graphs (Window Slicing, Window Warping, Feature Masking, Edge Perturbation, GCA), learnable graph augmentation (AD-GCL, Auto-GCL), supervised rs-fMRI models (STAGIN, Transformer), plus “No Pre-train” (same architecture without SSL) and a fully supervised pretrain (“Supervised”). pmc.ncbi.nlm.nih+1	On both REST-MDD and ABIDE, GCDA achieves the best accuracy/AUC among all methods; it improves over “No Pre-train” and “Supervised” baselines and over AD-GCL/Auto-GCL and classical augmentations, with statistically significant gains (p < 0.05) reported in tables comparing ACC/AUC/SEN/SPE/BAC. pmc.ncbi.nlm.nih+1
3	Chen et al., MICCAI 2024 – “Self-supervised Learning with Adaptive Graph Structure and Function Representation for Cross-Dataset Brain Disorder Diagnosis”	Self-supervised graph contrastive framework for cross-dataset diagnosis. A graph structure learner (GSL) learns subject-specific FC and EC graphs from BOLD signals; a multi-state brain network encoder (multi-state GIN) assigns features to multiple “brain states” and aggregates them with attention; pretext: contrastive loss between FCN and ECN views; downstream: fine-tune on new disease/dataset. pmc.ncbi.nlm.nih	Pretrain on 705 unlabeled ADNI rs-fMRI scans; cross-dataset transfer to ABIDE (ASD vs HC) and ADHD-200 (ADHD vs HC) with 5-fold CV on target datasets. pmc.ncbi.nlm.nih	Compared against BrainGB, BrainGSL (self-supervised), and UCGL. Also an ablation “Ours (w/o pretext)” that uses the same multi-state encoder but no SSL pretraining. pmc.ncbi.nlm.nih	In cross-dataset ADNI→ABIDE and ADNI→ADHD classification, the full method achieves higher ACC, AUC, SEN, SPE than BrainGB, BrainGSL, UCGL, and the no-pretext variant; improvements are statistically significant (p < 0.05) and on the order of a few percentage points in ACC/AUC. pmc.ncbi.nlm.nih
4	Chen et al., KBS 2025 – “Guiding fusion of dynamic functional and effective connectivity in spatio-temporal graph neural network for brain disorder classification”	Spatio-temporal GNN named DCSTN that fuses dynamic functional connectivity (dFC) and dynamic effective connectivity (dEC). Dynamic FC and EC sequences are built from rs-fMRI; a spatio-temporal GNN processes them and a “guiding” mechanism uses EC to guide fusion of FC representations for brain disorder classification. sciencedirect+1	KBS abstract/snippet states a general “brain disorder classification” setting with dynamic FC and EC; details of datasets and specific disorders are not visible in the accessible text (likely rs-fMRI, e.g. MCI/ADNI, but not explicitly stated in what we can see). sciencedirect+1	Abstract mentions DCSTN “outperforms state-of-the-art methods” and is a novel spatio-temporal GNN to fuse dynamic FC and EC, but does not list baselines or their names in the accessible snippet. sciencedirect+1	Reported to outperform unnamed SOTA methods for brain disorder classification when integrating dynamic FC and EC in feature space; no explicit numbers or baseline list are visible in the accessible abstract/snippet. sciencedirect+1
5	Hu et al., IEEE TMI 2024 – “Cost-sensitive weighted contrastive learning based on graph convolutional networks for imbalanced Alzheimer’s disease staging”	Cost-sensitive weighted contrastive learning method built on GCNs (CSWCL-GCN) for imbalanced multi-class AD staging using rs-fMRI. Construct a multi-view graph from FC and high-order FC (HOFC); introduce cost-sensitive weights and distance-to-prototype-based weights into the contrastive loss to emphasize minority classes and account for disease heterogeneity. pubmed.ncbi.nlm.nih	ADNI rs-fMRI, 720 subjects: 184 NC, 40 SMC, 208 EMCI, 172 LMCI, 116 AD. Multi-class staging task NC/SMC/EMCI/LMCI/AD. pubmed.ncbi.nlm.nih	Abstract states that CSWCL-GCN is compared with “state-of-the-art methods” on ADNI; specific baselines are not listed in the abstract we can access (likely prior FC/HOFC-GCN AD classification methods, but not named there). pubmed.ncbi.nlm.nih	Experimental results show CSWCL-GCN outperforming SOTA baselines on ADNI for imbalanced AD staging, improving classification performance especially on minority stages by using cost-sensitive weighted contrastive loss; exact numerical gains and baseline names are only available in the full article, not in the abstract. pubmed.ncbi.nlm.nih